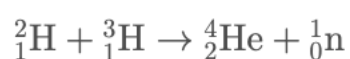


Section: Particles and Waves

Topic: Nuclear Reactions

A nuclear fusion reaction is represented by the equation:



(a)

Show that the total charge and total mass number are conserved in this reaction.

✔ Check **mass numbers** (top numbers):

$$2 + 3 = 5 \quad \text{on left,} \quad 4 + 1 = 5 \quad \text{on right}$$

✔ Check **charges** (bottom numbers):

$$1 + 1 = 2 \quad \text{on left,} \quad 2 + 0 = 2 \quad \text{on right}$$

✔ Both mass number and charge are conserved.

| ⚠ *Revision tip:* Always check nuclear equations for **conservation of charge and nucleon number** (protons + neutrons).

(b)

The mass of a deuterium nucleus is $3.343 \times 10^{-27} \text{ kg}$.

The mass of a tritium nucleus is $5.006 \times 10^{-27} \text{ kg}$.

The mass of the products is $8.374 \times 10^{-27} \text{ kg}$.

Calculate the energy released in the reaction.

Step 1: Find **mass difference** (Δm)

$$\Delta m = (3.343 + 5.006) - 8.374 = 8.349 - 8.374 = -0.025 \times 10^{-27} \text{ kg}$$

Take the magnitude:

$$\Delta m = 2.5 \times 10^{-29} \text{ kg}$$

Step 2: Use $E = mc^2$

$$E = 2.5 \times 10^{-29} \times (3.0 \times 10^8)^2 = 2.5 \times 9 \times 10^{-13} = 2.3 \times 10^{-12} \text{ J}$$

✔ *Revision tip:*

- Be careful with mass units — always in **kg**
- Fusion releases energy because **mass is lost and converted to energy**